INTERNSHIP NOTES • WEEK 1

PHP Form Submission & MySQL Database

PHP · MySQL · Bootstrap · HTML · JavaScript

Project Objective

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To collect user information through a web form and store it permanently in a MySQL database using PHP. The project demonstrates a complete full-stack workflow: HTML form creation with Bootstrap styling, server-side data handling with PHP, secure database connectivity, and persistent data storage via SQL INSERT operations.

Step 1 — Create the Database

A MySQL database was created to store user information submitted through the web form. The table was defined with the following columns:

Column	Type	Description
`id`	INT AUTO_INCREMENT	Primary key — auto-assigned unique identifier
`firstname`	VARCHAR(100)	User's first name
`lastname`	VARCHAR(100)	User's last name
`email`	VARCHAR(100)	User's email address
`language`	VARCHAR(100)	Preferred programming language

SQL — Create Database & Table

setup.sql

```
CREATE DATABASE users_db;
USE users_db;
CREATE TABLE users (
    id         INT AUTO_INCREMENT PRIMARY KEY,
    firstname  VARCHAR(100),
    lastname   VARCHAR(100),
    email      VARCHAR(100),
    language   VARCHAR(100)
);
```

Step 2 — Project Structure & Files

A project folder was created containing three PHP files, each with a distinct role in the application.

index.php	db.php	controller.php
HTML Form Layer	Database Connection	Form Processing
• HTML form (POST)	• Host / user / pass / db	• Reads $_POST data
• Bootstrap 5 styling	• mysqli_connect()	• Validates inputs
• 4 input fields	• Included on demand	• Executes INSERT query

db.php — Database Connection

db.php

```
<?php
$servername = "localhost";
$username   = "root";
$password   = "";
$database   = "users_db";
$conn = mysqli_connect($servername, $username, $password, $database);
if (!$conn) {
    die("Connection failed: " . mysqli_connect_error());
}
?>
```

index.php — HTML Form (condensed)

index.php

```
<!-- Bootstrap CSS -->
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
      rel="stylesheet">
<form action="controller.php" method="POST">
  <input type="text"  name="firstname" class="form-control" required>
  <input type="text"  name="lastname"  class="form-control" required>
  <input type="email" name="email"     class="form-control" required>
  <select name="language" class="form-select" required>
    <option value="PHP">PHP</option>
    <option value="JavaScript">JavaScript</option>
    <option value="Python">Python</option>
  </select>
  <button type="submit" class="btn btn-primary">Submit</button>
</form>
```

controller.php — Process & Insert

```php
<?php
include 'db.php';
if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $firstname = $_POST['firstname'];
    $lastname  = $_POST['lastname'];
    $email     = $_POST['email'];
    $language  = $_POST['language'];
    $sql = "INSERT INTO users (firstname, lastname, email, language)
           VALUES ('$firstname', '$lastname', '$email', '$language')";
    if (mysqli_query($conn, $sql)) {
        echo "Data inserted successfully!";
    } else {
        echo "Error: " . mysqli_error($conn);
    }
}
?>
```

Project Flow

The diagram below illustrates how data moves through the application from the user's browser to the MySQL database.

1	User opens index.php in the browser.
2	User fills in First Name, Last Name, Email, and Programming Language.
3	User clicks Submit — form data is sent to controller.php via HTTP POST.
4	controller.php reads the submitted values from $_POST.
5	controller.php includes db.php to establish the MySQL connection.
6	An INSERT query is executed to write the data into the users table.
7	A success or error message is displayed to confirm the operation.

Technologies Used

HTML5 Structure	**Bootstrap 5** Styling / Responsive UI	**JavaScript** Client-side behaviour
PHP Server-side logic	**MySQL** Relational database	